Teaching Guidelines for

Web Programming Technologies

PGCP-AC February 2026

Duration: 150 hours (60 theory hours + 60 lab hours +30 Self-Learning hrs)

Objective: To introduce the students to HTML, CSS, JavaScript, XML, JSON, Ajax, Node.js, Express.js, React, React-Redux,and practical relevance of all these technologies.

Evaluation: 100 marks
Weightage: CCEE – 40%, Lab exam – 40%, Internals – 20%

Text Books:
- Fundamentals of Web Development, 1e, by Randy Connolly, Ricardo Hoar / Pearson
- MERN Quick Start Guide – Build web applications with MongoDB, Express.js, React, and Node by Eddy Wilson IriarteKoroliova / Packt

References:
- Internet & World Wide Web : How to Program by Paul Deitel, Henry Deitel&Abbey Deitel / Pearson Education
- XML - How to Program by Deitelet al /Pearson Education
- Ajax in Action by Dave Crane, Eric Pascarello /Dreamtech Press
- JavaScript: The Good Parts by Douglas Crockford / O'Reilly
- Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and NodebyVasan Subramanian / Apress
- Web Application Security: A Beginner's Guide by Bryan Sullivan & Vincent Liu / Tata McGraw Hill
- W3Schools Tutorials [https://www.w3schools.com/]
- Mozilla Developer Network Web Development Tutorials [https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web]
- Curated Tutorial Links on ES6, React, etc. [https://github.com/markerikson/react-redux-links]

(Note: Each Session is of 2 hours)

Session 1: Architecture of Web (2T+ 2L)
Lecture:
- Brief history of the Internet
- How does the Internet work?
- Internet Protocol; HTTP
- Domain Names; Domain Name Service servers
- HTTP Protocols
  - Difference between HTTP1.0, HTTP 1.1, and HTTP 2.0
  - Methods – GET, POST, HEAD, PUT, DELETE, etc.
  - Status codes
  - Stateless nature of the protocol and HTTP Session
  - HTTPS
- Architecture of the Web
- Web servers – IIS, Apache server

Lab:

- Exploring different browsers
  - Mozilla Firefox, Google Chrome, Safari
- Exploring different text editors
  - Windows: Notepad++, Linux: Gedit or Vim or Emacs

Session 2: HTML (2T+ 2L)

Lecture:

- Introduction to HTML5
- Introduction to basic HTML Tags
  - Alignment, Headings, Anchor, Paragraph, Image, Lists, Tables, and iFrames
- HTML5
  - New features in HTML5
  - New elements, new attributes, link relations, microdata, ARIA accessibility, objects, events, and Canvas tags
  - HTML5 Validation
  - Audio & Video Support
  - Geo-location Support
- HTML Forms & Controls
  - Input, Text Area, Radio Button, Checkbox, Dropdown, Submit, Reset, Button, etc.
- Introduction to Document Object Model(DOM)

Lab:

- Create a HTML form for building your resume.

Session 3: Cascading Style Sheets (CSS) (2T+ 2L)

Lecture:

- Introduction to CSS, Styling HTML with CSS, Structuring pages with CSS,
- Inline CSS, Internal CSS, External CSS, Multiple styles, CSS Fonts
- CSS Box Model
- id Attribute, class Attribute
- HTML Style Tags
- Linking a style to an HTML document

Lab:

- Apply inline, internal,and external CSS to change colors of certain text portions, bold, underline, and italics certain words in the previously created HTML resume form.

Session 4: Responsive Web Design (2T+ 2L+ 3SL)

Lecture:

- Introduction of UI Scripting
- The Best Experience for All Users
  - Desktop, Tablet, Mobile
- Bootstrap
  - Overview of Bootstrap, Need to use Bootstrap
  - Bootstrap Grid System, Grid Classes, Basic Structure of a Bootstrap Grid
  - Typography
  - Components – Tables, Images, Jumbotron, Wells, Alerts, Buttons, Button Groups, Badges/Labels, Progress Bars, Pagination, List Groups, Panels, Dropdowns, Collapse, Tabs/Pills, Navbar
  - Forms, Inputs
  - Bootstrap Themes, Templates

Lab:

- Update the design of the Resume form using Bootstrap

Session 5: JavaScript (2T+ 2L)

Lecture:

- Introduction to JavaScript
- Variables in JavaScript
- Statements, Operators, Comments, Expressions, and Control Structures
- JavaScript Scopes
- Strings, String Methods
- Numbers, Number Methods
- Boolean Values
- Dates, Date Formats, Date Methods
- Arrays, Array Methods

Lab:

- Practice writing basic JavaScript programs for better understanding of the language constructs

Sessions 6 & 7: JavaScript (4T+ 4L)

Lecture:

- Objects, Object Definitions, Object Properties, Object Methods, Object Prototypes
- Functions, Function Definitions, Function Parameters, Function Invocation, Function Closures
- Introduction to Object Oriented Programming in JS
    - Method, Constructor, Inheritance, Encapsulation, Abstraction, Polymorphism

Lab:

- Write a JavaScript program to sort a list of elements by implementing a sorting algorithm.
- Write a JavaScript program to list the properties of a JavaScript object.

Sessions 8 & 9: JavaScript (4T+ 4L)

Lecture:

- Document Object Model (DOM)
    - Object hierarchy in JavaScript
    - HTML DOM, DOM Elements, DOM Events
    - DOM Methods, DOM Manipulation
- Forms, Forms API, Forms Validation
- Regular Expressions
- Errors, Debugging
- Introduction to Browser Dev Tool
- Pushing code quality via JSLint tool

Lab:

- Write a JavaScript function to get First and Last name from the previously created Resume form
- Validate the entire Resume form using client-side JavaScript
- Write a JavaScript function to validate whether a given value is RegEx or not.

Session 10: JSON & jQuery (2T+ 2L+ 3SL)

Lecture:

- JSON: JavaScript Object Notation (JSON)
    - Introduction and need of JSON
    - JSON Syntax Rules
    - JSON Objects, JSON Arrays, JSON Files

    - JSON parsing
- jQuery: Introduction
    - jQuery selectors
    - jQuery events
    - jQuery animation effects
    - jQuery DOM traversal and manipulation
    - Data attributes and templates
    - jQuery DOM utility functions
    - jQuery plugins

Lab:

- Create a JSON object, array, and file to store a cricket match (or any team sport) scoreboard.
- Write a jQuery program to get a single element from a selection of elements of a HTML page.
- You are having sample data for the link. Write jQuery code to change the hyperlink and the text of an existing link.
- Write a jQuery program to attach a click and double-click events to all <p> elements.
- Write a jQuery program to hide all headings on a page when they are clicked.
    - Also find the position of the mouse pointer relative to the left and top edges of the document.

Sessions 11 & 12: AJAX & Axios HTTP Client (4T+ 4L+ 3SL)

Lecture:

- AJAX: Asynchronous JavaScript and XML
    - Introduction to AJAX
    - AJAX framework and its architecture
    - Web services and AJAX
    - AJAX using jQuery and jQuery
- Axios: A promise-based HTTP client
    - The Axios instance and its config
    - Handling request and response
    - Handling errors

Lab:

- Design and implement a webpage that displays a live scoreboard. Use AJAX (XMLHttpRequest) to retrieve and interpret JSON data from a URL provided by the faculty.
- Design and implement a webpage that displays live news headlines. Use the Axios HTTP client to retrieve and interpret JSON data from a URL provided by the faculty.

Session 13: Introduction to Node.js (2T+ 2L+ 3SL)

Lecture:

- Introduction to Node.js
- Browser JS vs. Node.js
- ECMAScript 2015 (ES6)
- Node.js REPL

Lab:

- Install Node.js 12.x.x LTS version on your machine
- Write a recursive function in Node.js
- Write a Node program that prints all the numbers between 1 and 100, each on a separate line. A few caveats:
    - if the number is divisible by 3, print "foo"
    - if the number is divisible by 5, print "bar"
    - if the number is divisible by both 3 and 5, print "foobar"

Sessions 14 & 15: Node.js Asynchronous Programming (4T+ 4L)
Lecture:

- Introduction to Asynchronous programming and callbacks
- Promises and async& await
- The Event Loop and Timers

Lab:

- Assignment on JavaScript callback functions
- Assignment on Timers, Promises, and Async& Await

Session 16: Node.js Modules(2T+ 2L)
Lecture:

- Understanding Node modules, exports, and require
- Introduction to npm
  - package.json and package-lock.json files
  - Install, update, and manage package dependencies
  - Local and global packages

Lab:

- Create a module and import it in other programs
- Install a module/package using npm

Session 17: Node.js Modules – *fs* and *http* (2T+ 2L+ 3SL)
Lecture:

- File I/O – Sync &Async Methods
- HTTP Module – Building an HTTP server
- Developing a Node web application

Lab:

- Write a program to create a new file and write some content to it in synchronous mode and read and display file contents on standard output in async mode
- Build a simple Node.js web application serving both HTTP GET and POST methods

Session 18: Introduction to Express (2T+ 2L+ 3SL)
Lecture:

- Introduction to Express
- Getting started with Express
- Application, Request and Response Objects
- Routes and Middlewares
- Templates, Template Engines, and Rendering Views

Lab:

- Use Node and Express to write a simple web application that consists of at least 5 route implementations
- Rebuild any previous Node assignment using Express and a template engine

Session 19: Introduction to React (2T+ 2L+ 3SL)
Lecture:

- Introduction to React
- Getting started with React
- React Elements and React Components
- Function and Class Components
- Working with React Components and Props
  - Compose components

  - Render components
  - Declutter components

Lab:

- Rebuild any previous plain HTMLlabassignment using React
- Build a React Clock app showing time (hh:mm:ss) of any three countries

Sessions 20, 21 & 22: React (6T+ 6L+ 3SL)

Lecture:

- Introduction to State and Lifecycle
- Statefulcomponents and lifecycle methods
- Props vs. State vs. Context
- Handling events
- Conditional rendering

Lab:

- Implement the following items in theReact Clock app
  - Update the time (hh:mm:ss) using State and Lifecycle methods
  - Add a close functionon each rendered clock component
  - Assign background color of rendered clock components based on AM, PM

Session 23, 24 & 25: React (6T+ 6L+ 3SL)

Lecture:

- Lists and Keys
  - Rendering Multiple Components
  - Basic List Component
- Working with forms and inputs
- Refs and the DOM
- Lifting state up

Lab:

- Implement and integrate a new feature in the React Clock app where one can select a country time zone from dropdown list and click on “Add” button to render it.

Session 26, 27: React (4T+ 4L)

Lecture:

- Error Boundaries
- Composition vs. Inheritance
  - Containment
  - Specialization
- Thinking in React

Lab:

- Implement error boundaries at appropriate places in the React Clock app

Session 28, 29 & 30: React-Redux (6T+ 6L+ 3SL)

Lecture:

- Introduction to Redux
- Actions, Reducers, and Stores
- Usage with React

Lab:

- Make necessary changes in the design and implementation of React Clock app using React-Redux to maintain the application state.